

On efficient quantum block encoding of pseudo-differential operators

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Abstract

Block encoding is a fundamental subroutine in various quantum algorithms with exponential speedup potential. In this paper, based on the framework introduced by Li et al.[1], we study the efficient quantum block encoding of pseudo-differential operators (PDOs), a ubiquitous class of dense operators in scientific computing. We first review the systematic construction of block encodings for PDOs with generic symbols, and then discuss efficiency improvements when the symbols possess separable or fully separable structures. Furthermore, as an original contribution to optimize the circuit depth, we propose an improved block encoding scheme for fully separable PDOs by incorporating the Approximate Quantum Fourier Transform (AQFT). We rigorously prove that substituting the exact QFT with the AQFT reduces the gate complexity bottleneck of the Fourier transforms from $\mathcal{O}(p^2d)$ to $\mathcal{O}\left(pd \log\left(\frac{pd}{\varepsilon}\right)\right)$ without violating the overall error tolerance. This improvement makes the quantum simulation of high-dimensional PDOs significantly more scalable.

1 Introduction

In this paper, we consider the problem of block encoding a large family of dense operators: the pseudo-differential operators (PDOs). PDO is a rich family of linear operators that include many commonly used examples in scientific problems, which is typically given in the following form:

$$Af(\mathbf{x}) = \int_{\mathbb{R}^d} e^{2\pi i \mathbf{x} \cdot \boldsymbol{\xi}} a(\mathbf{x}, \boldsymbol{\xi}) \widehat{f}(\boldsymbol{\xi}) d\boldsymbol{\xi}, \quad (1)$$

where $a(\mathbf{x}, \boldsymbol{\xi}) \in C^\infty(\mathbb{R}^d \times \mathbb{R}^d)$ is called the **symbol** of A and $\widehat{f}$ is the Fourier transform of f .

The PDOs considered in this paper are equipped with a periodic boundary condition on the space domain $\Omega = [0, 1]^d$. The frequency variable $\boldsymbol{\xi}$ thus takes the value on the integer grid, and the operator A becomes

$$Af(\mathbf{x}) = \sum_{\boldsymbol{\xi} \in \mathbb{Z}^d} e^{2\pi i \mathbf{x} \cdot \boldsymbol{\xi}} a(\mathbf{x}, \boldsymbol{\xi}) \widehat{f}(\boldsymbol{\xi}), \quad (2)$$

where $\widehat{f}$ is the coefficient of the Fourier series of f .

In this paper, we derive block encoding schemes for the PDO (2) based on different additional structures of the symbol $a(\mathbf{x}, \boldsymbol{\xi})$ [1].

2 Notations and preparation

2.1 Notations

We adopt the commonly used notation for binary numbers: for an integer power of two $P = 2^p$ and any $y \in [P]$ if

$$y = y_0 + 2y_1 + \cdots + 2^{p-1}y_{p-1} = (y_{p-1}y_{p-2} \cdots y_0)$$

in the binary system, the corresponding quantum state is $|y\rangle \equiv |y_{p-1} \cdots y_1 y_0\rangle$.

This extends to an m -tuple $\mathbf{x} = (x_1, \dots, x_m)$ with $x_j \in [P]$. The corresponding quantum state is given by $|x_m\rangle \cdots |x_1\rangle$, where $|x_j\rangle = |x_{j,p-1} \cdots x_{j,0}\rangle, \forall j$.

For a multivariate function $g : [P]^m \rightarrow \mathbb{R}$, we denote by D_g the diagonal multiplication operator on $\mathbb{C}^{mp}$:

$$D_g : |x_m\rangle \cdots |x_1\rangle \mapsto g(x_1, \dots, x_m) |x_m\rangle \cdots |x_1\rangle. \quad (3)$$

For a vector $\mathbf{v} = (v_1, v_2, \dots, v_m)$, we denote by $\text{diag}(\mathbf{v})$ the diagonal matrix $\text{diag}(v_1, v_2, \dots, v_m)$.

2.2 Discretization of pseudo-differential operators

As mentioned in [Section 1](#), the PDO considered in this paper is defined for periodic functions on $\Omega = [0, 1]^d$:

$$Af(\mathbf{x}) = \sum_{\boldsymbol{\xi} \in \mathbb{Z}^d} e^{2\pi i \mathbf{x} \cdot \boldsymbol{\xi}} a(\mathbf{x}, \boldsymbol{\xi}) \widehat{f}(\boldsymbol{\xi}).$$

In most numerical treatments, the function f is given on a discrete grid

$$X = \left\{ \frac{\mathbf{x}}{P} \equiv \left(\frac{x_1}{P}, \dots, \frac{x_d}{P} \right) : x_j \in [P], \forall j \right\},$$

where $P = 2^p$ is the number of discrete points used for each dimension, and the frequency domain is discretized correspondingly on $\left\{ -\frac{P}{2}, \dots, \frac{P}{2} - 1 \right\}^d$, which leads to the discretized PDO:

$$Af(\mathbf{x}) \equiv \sum_{\boldsymbol{\xi} \in \left\{ -\frac{P}{2}, \dots, \frac{P}{2} - 1 \right\}^d} e^{2\pi i \mathbf{x} \cdot \boldsymbol{\xi} / P} a\left(\frac{\mathbf{x}}{P}, \boldsymbol{\xi}\right) \widehat{f}(\boldsymbol{\xi}), \quad \mathbf{x} \in [P]^d. \quad (4)$$

Though the frequency variable $\boldsymbol{\xi}$ is discretized on $\left\{ -\frac{P}{2}, \dots, \frac{P}{2} - 1 \right\}^d$ in (4), by the convention of DFT, the frequency $\left(\frac{P}{2}, \dots, P - 1 \right)$ is often identified with $\left(-\frac{P}{2}, \dots, -1 \right)$, respectively, since P is a period for the frequency variable after DFT. In other words, the discretized PDO can be written as

$$Af(\mathbf{x}) = \sum_{\boldsymbol{\xi} \in [P]^d} e^{2\pi i \mathbf{x} \cdot \boldsymbol{\xi} / P} \tilde{a}\left(\frac{\mathbf{x}}{P}, \boldsymbol{\xi}\right) \widehat{f}(\boldsymbol{\xi}), \quad \mathbf{x} \in [P]^d,$$

where

$$\tilde{a}(\mathbf{x}, \boldsymbol{\xi}) \equiv a\left(\mathbf{x}, \boldsymbol{\xi} - P \sum_{\xi_j \geq P/2} \mathbf{e}_j\right),$$

and $\mathbf{e}_j$ is the j -th standard basis vector in $\mathbb{C}^d$.

To simplify the notation and avoid repetitive use of P , we further define

$$\check{a}(\mathbf{x}, \boldsymbol{\xi}) \equiv \tilde{a}\left(\frac{\mathbf{x}}{P}, \boldsymbol{\xi}\right),$$

and the discretized PDO becomes

$$Af(\mathbf{x}) = \sum_{\boldsymbol{\xi} \in [P]^d} e^{2\pi i \mathbf{x} \cdot \boldsymbol{\xi} / P} \check{a}(\mathbf{x}, \boldsymbol{\xi}) \widehat{f}(\boldsymbol{\xi}), \quad \mathbf{x} \in [P]^d. \quad (5)$$

It is clear that $\sup|\check{a}| = \sup|a|$. We also refer to $\check{a}$ as the symbol of the PDO to be computed.

2.3 Quantum Subroutines for Phase Multiplications and Diagonal Operators

Before discussing the block encoding of PDOs, we establish several essential quantum subroutines that will be used repeatedly to implement phase multiplications and diagonal operators.

First, we develop an efficient implementation for phase multiplication without involving any extra error or ancilla qubits.

Lemma 2.1. *The $2p$ -qubit circuit U_{ph} displayed in Figure 1 implements the unitary operator:*

$$|x\rangle |\xi\rangle \mapsto e^{2\pi i x \xi / P} |x\rangle |\xi\rangle, \quad x, \xi \in [P],$$

with $\mathcal{O}(p^2)$ gate complexity precisely without ancilla qubits.

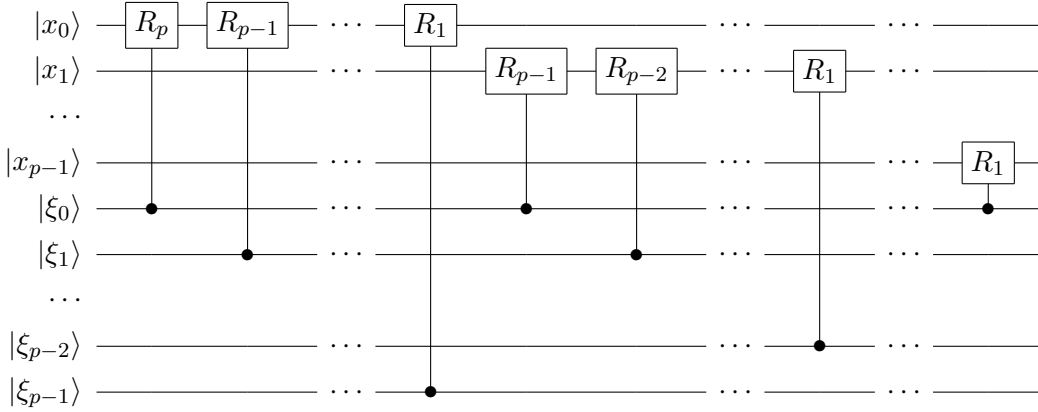


Figure 1: Circuit for phase multiplication U_{ph} .

Proof. We first write the binary representation of integers $x = (x_{p-1} \cdots x_0)$, $\xi = (\xi_{p-1} \cdots \xi_0)$, and do the following calculation:

$$\begin{aligned} & e^{2\pi i x \xi / P} |x\rangle |\xi\rangle \\ &= \prod_{j=0}^{p-1} \prod_{k=0}^{p-1} e^{2\pi i x_j \xi_k \cdot 2^{j+k-p}} |x_{p-1} \cdots x_0\rangle |\xi_{p-1} \cdots \xi_0\rangle \\ &= \prod_{0 \leq j+k < p} e^{2\pi i x_j \xi_k \cdot 2^{j+k-p}} |x_{p-1} \cdots x_0\rangle |\xi_{p-1} \cdots \xi_0\rangle \\ &= \left(\left(\prod_{k=0}^{p-1} e^{2\pi i x_0 \xi_k \cdot 2^{k-p}} \right) |x_0\rangle \right) \otimes \cdots \otimes \left(\left(\prod_{k=0}^0 e^{2\pi i x_{p-1} \xi_k \cdot 2^{k-1}} \right) |x_{p-1}\rangle \right) \otimes |\xi_{p-1} \cdots \xi_0\rangle. \end{aligned} \tag{6}$$

The circuit corresponding to the unitary in (6) can be implemented by a series of controlled rotations

$$\begin{array}{ccc} |x_j\rangle & \xrightarrow{R_{p-j-k}} & e^{2\pi i x_j \xi_k \cdot 2^{j+k-p}} |x_j\rangle \\ |\xi_k\rangle & \xrightarrow{\bullet} & |\xi_k\rangle \end{array}$$

where $R_j = R_z(\pi/2^{j-1})$. Finally, the circuit shown in Figure 1 is obtained after arranging the controlled rotations in the corresponding places. $\square$

Next, we state a general result for an efficient block encoding of diagonal arithmetic matrices D_g .

Proposition 2.2. *Assume that $g : \mathbb{R}^m \rightarrow \mathbb{R}$ is a smooth arithmetic function with $\sup |g| < +\infty$. Then there is an $(C, \mathcal{O}(\text{polylog}(1/\varepsilon) + \text{poly}(mp)), \varepsilon)$ -block-encoding of D_g with $\mathcal{O}(\text{polylog}(1/\varepsilon) + \text{poly}(mp))$ gates, where D_g is a diagonal operator on $\mathbb{C}^{mp}$ defined in (3), and $C \geq \sup |g|$.*

Proof. The circuit U_g is constructed as follows. Let $t = \lceil \log_2 \left(\frac{C\pi}{\varepsilon} \right) \rceil$, and let $\theta(x_1, \dots, x_m)$ be a map that gives $|\theta_{\text{sgn}} \theta_{t-1} \theta_{t-2} \dots \theta_0\rangle$, where $(\theta_{t-1} \theta_{t-2} \dots \theta_0)$ is the closest t -bit fixed-point representation of $\frac{1}{\pi} \arcsin \left(\frac{1}{C} |g(x_1, \dots, x_m)| \right)$, and $\theta_{\text{sgn}} = \mathbb{1}_{g < 0}$. For an arbitrary basis state $|x_m\rangle \dots |x_1\rangle$, we consider the system with $t + 2$ ancilla qubits $|0\rangle |x_m\rangle \dots |x_1\rangle |0\rangle^{\otimes t+1}$. Using the reversible computational model and uncomputation, the classical circuit:

$$|0\rangle |x_m\rangle \dots |x_1\rangle |0\rangle^{\otimes t+1} \mapsto |0\rangle |x_m\rangle \dots |x_1\rangle |\theta_{\text{sgn}} \theta_{t-1} \theta_{t-2} \dots \theta_0\rangle$$

can be constructed with $\mathcal{O}(\text{poly}(t) + \text{poly}(mp))$ gates and $\mathcal{O}(\text{poly}(mp))$ ancilla qubits. Then we apply the circuit in Figure 2 on the $t + 2$ ancilla qubits.

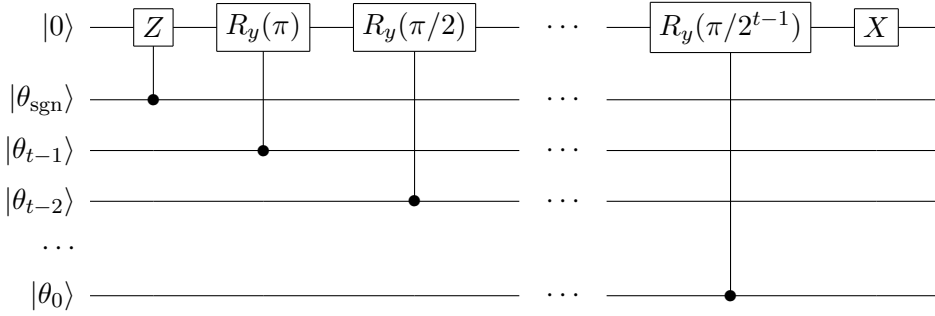


Figure 2: The controlled rotation part of U_g .

The state obtained is:

$$(-1)^{\theta_{\text{sgn}}} (\cos(\pi(\theta_{t-1} \dots \theta_0)) |1\rangle + \sin(\pi(\theta_{t-1} \dots \theta_0)) |0\rangle) |x_m\rangle \dots |x_1\rangle |\theta_{\text{sgn}} \theta_{t-1} \theta_{t-2} \dots \theta_0\rangle,$$

which can then be mapped to

$$(-1)^{\theta_{\text{sgn}}} (\cos(\pi(\theta_{t-1} \dots \theta_0)) |1\rangle + \sin(\pi(\theta_{t-1} \dots \theta_0)) |0\rangle) |x_m\rangle \dots |x_1\rangle |0\rangle^{\otimes t+1},$$

via uncomputation. Since

$$\left| (\theta_{t-1} \dots \theta_0) - \frac{1}{\pi} \arcsin \left(\frac{1}{C} |g(x_1, \dots, x_m)| \right) \right| < \frac{\varepsilon}{C\pi},$$

we have

$$\left| (-1)^{\theta_{\text{sgn}}} \sin(\pi(\theta_{t-1} \dots \theta_0)) - \frac{1}{C} g(x_1, \dots, x_m) \right| < \frac{\varepsilon}{C}$$

which means the desired state $\frac{1}{C} g(x_1, \dots, x_m) |x_m\rangle \dots |x_1\rangle$ is obtained with error at most $\frac{\varepsilon}{C}$ upon measuring the ancilla qubits and getting $|0\rangle^{\otimes t+2}$. $\square$

To further support the block encoding via Quantum Eigenvalue Transformation (QET), we introduce a lemma that provides Hermitian block encodings for two diagonal multiplication prototypes.

Lemma 2.3. For a fixed $P = 2^p$, let $\mathbf{v}_-, \mathbf{v}_+$ be the vectors:

$$\mathbf{v}_- = \left(0, 1, \dots, \frac{P}{2} - 1, -\frac{P}{2}, -\frac{P}{2} + 1, \dots, -1 \right), \quad \mathbf{v}_+ = \left(0, 1, \dots, \frac{P}{2} - 1, \frac{P}{2}, \frac{P}{2} + 1, \dots, P - 1 \right),$$

respectively, and D_-, D_+ be the diagonal matrices $\text{diag}(\mathbf{v}_-), \text{diag}(\mathbf{v}_+)$, respectively. Then there is an $(1, 1)$ -Hermitian-block-encoding of $\sin(\theta D_\sigma)$ with gate complexity $2p + 5$, where $\sigma \in \{-, +\}$ and $\theta > 0$ is a parameter.

Proof. For an arbitrary parameter $\theta > 0$, we first construct matrices $R_- = e^{i\theta D_-}$ and $R_+ = e^{i\theta D_+}$ with quantum circuits. Using the binary representation of ξ , we get

$$R_- |\xi_{p-1} \cdots \xi_0\rangle = e^{i((-\xi_{p-1})\xi_{p-2} \cdots \xi_0)\theta} |\xi_{p-1} \cdots \xi_0\rangle, \quad R_+ |\xi_{p-1} \cdots \xi_0\rangle = e^{i(\xi_{p-1} \cdots \xi_0)\theta} |\xi_{p-1} \cdots \xi_0\rangle.$$

Then

$$\begin{aligned} R_+ |\xi_{p-1} \cdots \xi_0\rangle &= e^{i(\xi_{p-1} \cdots \xi_0)\theta} |\xi_{p-1} \cdots \xi_0\rangle = e^{i\sum_{j \in [p]} \xi_j 2^j \theta} |\xi_{p-1} \cdots \xi_0\rangle \\ &= \prod_{j \in [p]} e^{i\xi_j 2^j \theta} \bigotimes_{j \in [p]} |\xi_j\rangle = \bigotimes_{j \in [p]} e^{i\xi_j 2^j \theta} |\xi_j\rangle = \bigotimes_{j \in [p]} R_z(2^j \theta) |\xi_j\rangle, \end{aligned}$$

and

$$R_- |\xi_{p-1} \cdots \xi_0\rangle = e^{-i\xi_{p-1} 2^p \theta} R_+ |\xi_{p-1} \cdots \xi_0\rangle = \bigotimes_{j \in [p]} R_z((-1)^{\delta_{j,p-1}} 2^j \theta) |\xi_j\rangle,$$

so $R_+ = \bigotimes_{j \in [p]} R_z(2^j \theta)$ and $R_- = \bigotimes_{j \in [p]} R_z((-1)^{\delta_{j,p-1}} 2^j \theta)$. Let U_σ be the circuit displayed in Figure 3, where $\sigma \in \{-, +\}$, then U_σ is a $(1, 1)$ -Hermitian-block-encoding of $\sin(\theta D_\sigma)$.

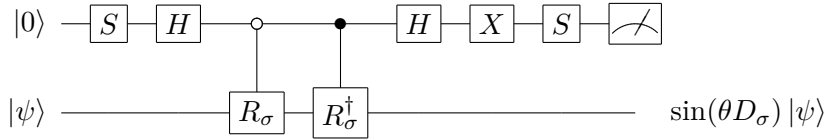


Figure 3: Hermitian block encoding of $\sin(\theta D_\sigma)$.

In fact, the matrix corresponding to U_σ is

$$\begin{aligned} U_\sigma &= \begin{bmatrix} I & \\ & iI \end{bmatrix} \begin{bmatrix} I & \\ & I \end{bmatrix} \frac{1}{\sqrt{2}} \begin{bmatrix} I & I \\ I & -I \end{bmatrix} \begin{bmatrix} R_\sigma & \\ & R_\sigma^\dagger \end{bmatrix} \frac{1}{\sqrt{2}} \begin{bmatrix} I & I \\ I & -I \end{bmatrix} \begin{bmatrix} I & \\ & iI \end{bmatrix} \\ &= \frac{1}{2} \begin{bmatrix} R_\sigma - R_\sigma^\dagger & i(R_\sigma + R_\sigma^\dagger) \\ i(R_\sigma + R_\sigma^\dagger) & -(R_\sigma - R_\sigma^\dagger) \end{bmatrix} = i \begin{bmatrix} \sin(\theta D_\sigma) & \cos(\theta D_\sigma) \\ \cos(\theta D_\sigma) & -\sin(\theta D_\sigma) \end{bmatrix}, \end{aligned}$$

and this is a Hermitian matrix with the first diagonal block being $\sin(\theta D_\sigma)$ after ignoring the global phase factor i . It can be seen from Figure 3 that the number of elementary gates used is $2p + 5$. $\square$

By restricting the parameter θ to be $0 < \theta < \frac{\pi}{2P}$ to recover θD_σ from $\sin(\theta D_\sigma)$ with the arcsin function, one obtains the following block encoding proposition based on QET.

Proposition 2.4. *Let U_- and U_+ be the $(1, 1)$ -Hermitian-block-encodings of $\sin(\theta D_-)$ and $\sin(\theta D_+)$ constructed in Lemma 2.3 for $0 < \theta < \frac{\pi}{2P}$ with $P = 2^p$. Assume that g is an even (resp. odd) continuous real function on $[-P, P]$ and that $\deg_g(\varepsilon)$ is the smallest positive integer such that there exists an even (resp. odd) polynomial $\tilde{g}$ with the degree bounded by $\deg_g(\varepsilon)$ satisfying*

$$\sup_{|x| \leq \sin(P\theta)} \left| C_g \tilde{g}(x) - g\left(\frac{1}{\theta} \arcsin x\right) \right| < \varepsilon, \quad \sup_{|x| \leq 1} |\tilde{g}(x)| \leq 1, \quad (7)$$

where $C_g \geq \sup |g|$ is a constant. Then there is a $(C_g, 2, \varepsilon)$ -block-encoding for both $g(D_-)$ and $g(D_+)$ with $\mathcal{O}(p \deg_g(\varepsilon))$ gates, where D_- and D_+ are defined in Lemma 2.3.

Proof. The QET circuit with U_σ gives a $(C_g, 2)$ -block-encoding for $\tilde{g}(\sin(\theta D_\sigma))$, and since

$$\sup_{|x| \leq \sin(P\theta)} \left| C_g \tilde{g}(x) - g\left(\frac{1}{\theta} \arcsin x\right) \right| < \varepsilon,$$

we have

$$\left\| C_g \tilde{g}(\sin(\theta D_\sigma)) - g\left(\frac{1}{\theta} \arcsin(\sin(\theta D_\sigma))\right) \right\| < \varepsilon,$$

Thus the QET circuit with U_σ gives a $(C_g, 2, \varepsilon)$ -block-encoding for

$$g\left(\frac{1}{\theta} \arcsin(\sin(\theta D_\sigma))\right) = g\left(\frac{1}{\theta} \cdot \theta D_\sigma\right) = g(D_\sigma),$$

where $\sigma \in \{-, +\}$ and we have used the fact that $0 < \theta < \frac{\pi}{2P}$ in the first equality. Since $\mathcal{O}(p)$ gates are used in U_σ , the gate complexity of the circuit described above is $\mathcal{O}(p \deg_g(\varepsilon))$. $\square$

The gate complexity of the circuit built with QET in [Proposition 2.4](#) depends on the smoothness of g . For instance, we have the following corollary.

Corollary 2.5. *Assume that g is an even (resp. odd) differentiable real function on $[-\frac{\pi}{2\theta}, \frac{\pi}{2\theta}]$. Let $\eta_g(\varepsilon, \theta)$ be the smallest integer such that there exists a polynomial u with degree $\eta_g(\varepsilon, \theta)$ satisfying*

$$\sup_{|y| < \frac{\pi}{2\theta}} |u(y) - g(y)| < \frac{\varepsilon}{3},$$

then the gate complexity of the circuit used in [Proposition 2.4](#) is

$$\mathcal{O}\left(p \log\left(\frac{C'_g}{\theta \varepsilon}\right) \eta_g(\varepsilon, \theta)\right),$$

where $C'_g = \sup_{|y| < \frac{\pi}{2\theta}} |g'(y)|$.

Proof. Without loss of generality, assume $1 > \frac{\varepsilon}{2C'_g}$, otherwise we can just let $\tilde{g} = 0$ in (7). Since $\frac{1}{\theta} \arcsin(x)$ is an analytic function, there is a truncation v of the Taylor series of $\frac{1}{\theta} \arcsin(x)$ with degree $\mathcal{O}\left(\log\left(\frac{C'_g}{\theta \varepsilon}\right)\right)$ such that

$$\sup_{|x| < \sin(P\theta)} \left| v(x) - \frac{1}{\theta} \arcsin(x) \right| < \frac{\varepsilon}{3C'_g}.$$

Now since the coefficients of the Taylor series of $\frac{1}{\theta} \arcsin(x)$ at $x = 0$ are all non-negative, it holds that $v([-1, 1]) \subset [-\frac{\pi}{2\theta}, \frac{\pi}{2\theta}]$. Let $\bar{g}(x) = \frac{1}{C'_g} u(v(x))$ and $\tilde{g} = (1 - \frac{\varepsilon}{3C'_g})\bar{g}$, then $\tilde{g}$ has degree

$$\mathcal{O}\left(\log\left(\frac{C'_g}{\theta \varepsilon}\right) \eta_g(\varepsilon, \theta)\right),$$

and $\forall |x| \leq \sin(P\theta)$,

$$\begin{aligned} \left| C_g \bar{g}(x) - g\left(\frac{1}{\theta} \arcsin(x)\right) \right| &\leq |u(v(x)) - g(v(x))| + \left| g(v(x)) - g\left(\frac{1}{\theta} \arcsin(x)\right) \right| \\ &< \frac{\varepsilon}{3} + \sup_{|y| < \frac{\pi}{2\theta}} |g'(y)| \left| v(x) - \frac{1}{\theta} \arcsin(x) \right| < \frac{\varepsilon}{3} + C'_g \cdot \frac{\varepsilon}{3C'_g} = \frac{2\varepsilon}{3}. \end{aligned}$$

In addition, since v maps $[-1, 1]$ into $[-\frac{\pi}{2\theta}, \frac{\pi}{2\theta}]$, we have

$$|\bar{g}(x)| \leq \sup_{|y| < \frac{\pi}{2\theta}} \left| \frac{1}{C_g} u(y) \right| \leq \sup_{|y| < \frac{\pi}{2\theta}} \left| \frac{1}{C_g} u(y) - \frac{1}{C_g} g(y) \right| + \sup_{|y| < \frac{\pi}{2\theta}} \left| \frac{1}{C_g} g(y) \right| \leq \frac{\varepsilon}{3C_g} + 1,$$

and therefore $|\tilde{g}(x)| = (1 - \frac{\varepsilon}{3C_g})|\bar{g}(x)| < 1, \forall x \in [-1, 1]$. In addition, we have $\forall |x| \leq \sin(P\theta)$,

$$\begin{aligned} \left| C_g \tilde{g}(x) - g\left(\frac{1}{\theta} \arcsin(x)\right) \right| &= \left| \left(1 - \frac{\varepsilon}{3C_g}\right) \left(C_g \bar{g}(x) - g\left(\frac{1}{\theta} \arcsin(x)\right) \right) + \frac{\varepsilon}{3C_g} g\left(\frac{1}{\theta} \arcsin(x)\right) \right| \\ &\leq \left(1 - \frac{\varepsilon}{3C_g}\right) \frac{2\varepsilon}{3} + \frac{\varepsilon}{3C_g} C_g < \frac{2\varepsilon}{3} + \frac{\varepsilon}{3} = \varepsilon. \end{aligned}$$

According to [Proposition 2.4](#), we have

$$\deg_g(\varepsilon) = \mathcal{O} \left(\log \left(\frac{C'_g}{\theta \varepsilon} \right) \eta_g(\varepsilon, \theta) \right)$$

and the gate complexity of the circuit used in [Proposition 2.4](#) is

$$\mathcal{O} \left(p \log \left(\frac{C'_g}{\theta \varepsilon} \right) \eta_g(\varepsilon, \theta) \right),$$

where the factor p comes from preparing $\sin(\theta D_\sigma)$ as mentioned in [Lemma 2.3](#). □

3 Block encoding for generic symbols

This section is concerned with the block encoding of the PDO (5) with a generic symbol $a(\mathbf{x}, \boldsymbol{\xi})$, without assuming any additional structure. In order to compute the PDO in (5), a simple strategy is to first lift the state to the phase space $[P]^d \times [P]^d$. Then the multiplication of $a(\mathbf{x}, \boldsymbol{\xi})$ in (5) can be performed by diagonal matrix block encodings. We can construct the entire circuit as illustrated in [Figure 4](#).

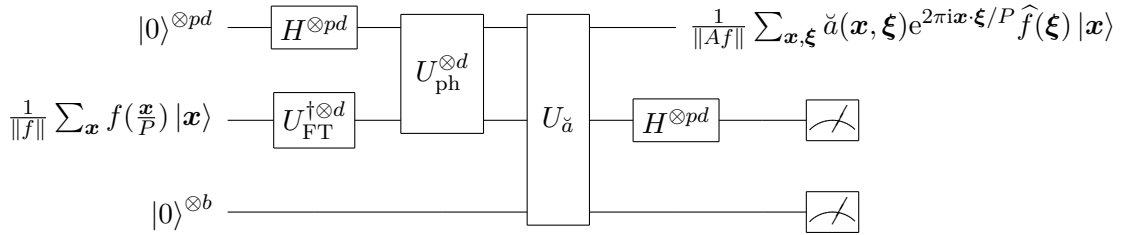


Figure 4: Circuit that implements the PDO with a generic symbol.

Now we explain the circuit displayed in [Figure 4](#) in more details. First, we assume the accessibility of the state $\frac{1}{\|f\|} \sum_{\mathbf{x}} f(\frac{\mathbf{x}}{P}) |\mathbf{x}>$ as an input, where

$$\|f\| = \sqrt{\sum_{\mathbf{x} \in [P]^d} \left| f\left(\frac{\mathbf{x}}{P}\right) \right|^2}$$

is the normalization factor.

Step 1. Apply QFT and lift the input state to the phase space. We first obtain the representation of f in the frequency domain by QFT. After applying the IQFT to the state $\frac{1}{\|f\|} \sum_{\mathbf{x}} f(\frac{\mathbf{x}}{P}) |\mathbf{x}\rangle$ for d times, we get

$$\frac{1}{\|f\|} \frac{1}{\sqrt{P^d}} \sum_{\boldsymbol{\xi}} \sum_{\mathbf{x}} f\left(\frac{\mathbf{x}}{P}\right) e^{-2\pi i \boldsymbol{\xi} \cdot \mathbf{x}/P} |\boldsymbol{\xi}\rangle = \frac{\sqrt{P^d}}{\|f\|} \sum_{\boldsymbol{\xi}} \hat{f}(\boldsymbol{\xi}) |\boldsymbol{\xi}\rangle.$$

Then the state $\frac{1}{\|f\|} \sum_{\mathbf{x}, \boldsymbol{\xi}} \hat{f}(\boldsymbol{\xi}) |\mathbf{x}\rangle |\boldsymbol{\xi}\rangle$ is obtained by applying the Hadamard gates $H^{\otimes pd}$ to the $\mathbf{x}$ -register and putting both registers together.

Step 2. Multiply the phase $e^{2\pi i \mathbf{x} \cdot \boldsymbol{\xi}/P}$ with U_{ph} . Rewriting $\frac{1}{\|f\|} \sum_{\mathbf{x}, \boldsymbol{\xi}} e^{2\pi i \mathbf{x} \cdot \boldsymbol{\xi}/P} \hat{f}(\boldsymbol{\xi}) |\mathbf{x}\rangle |\boldsymbol{\xi}\rangle$ as

$$\frac{1}{\|f\|} \sum_{\mathbf{x}, \boldsymbol{\xi}} e^{2\pi i x_d \xi_d/P} \dots e^{2\pi i x_1 \xi_1/P} \hat{f}(\boldsymbol{\xi}) |\mathbf{x}\rangle |\boldsymbol{\xi}\rangle,$$

then the map from $\frac{1}{\|f\|} \sum_{\mathbf{x}, \boldsymbol{\xi}} \hat{f}(\boldsymbol{\xi}) |\mathbf{x}\rangle |\boldsymbol{\xi}\rangle$ to $\frac{1}{\|f\|} \sum_{\mathbf{x}, \boldsymbol{\xi}} e^{2\pi i \mathbf{x} \cdot \boldsymbol{\xi}/P} \hat{f}(\boldsymbol{\xi}) |\mathbf{x}\rangle |\boldsymbol{\xi}\rangle$ can be performed by applying the phase multiplication circuit U_{ph} established in Lemma 2.1 for d times to the register pairs $(x_d, \xi_d), \dots, (x_1, \xi_1)$. The corresponding circuit is denoted as $U_{\text{ph}}^{\otimes d}$ and involves $\mathcal{O}(p^2 d)$ elementary gates with no ancilla qubits.

Step 3. Multiply the symbol $\check{a}(\mathbf{x}, \boldsymbol{\xi})$. The next component in Figure 4 is the diagonal multiplication $U_{\check{a}}$, which is designed to approximate the map

$$\frac{1}{\|f\|} \sum_{\mathbf{x}, \boldsymbol{\xi}} e^{2\pi i \mathbf{x} \cdot \boldsymbol{\xi}/P} \hat{f}(\boldsymbol{\xi}) |\mathbf{x}\rangle |\boldsymbol{\xi}\rangle |0\rangle^{\otimes b} \mapsto \frac{1}{C_a \|f\|} \sum_{\mathbf{x}, \boldsymbol{\xi}} \check{a}(\mathbf{x}, \boldsymbol{\xi}) e^{2\pi i \mathbf{x} \cdot \boldsymbol{\xi}/P} \hat{f}(\boldsymbol{\xi}) |\mathbf{x}\rangle |\boldsymbol{\xi}\rangle |0\rangle^{\otimes b} + |\perp\rangle,$$

where $C_a > 0$ is a constant that depends on $\check{a}(\mathbf{x}, \boldsymbol{\xi})$ and b is the number of ancilla qubits used for $U_{\check{a}}$. As a result of applying the arithmetic block encoding given in Proposition 2.2 with $m = 2d, g = \check{a}$ and ε replaced by $\frac{\varepsilon}{\sqrt{P^d}}$ to the state $\frac{1}{\|f\|} \sum_{\mathbf{x}, \boldsymbol{\xi}} e^{2\pi i \mathbf{x} \cdot \boldsymbol{\xi}/P} \hat{f}(\boldsymbol{\xi}) |\mathbf{x}\rangle |\boldsymbol{\xi}\rangle |0\rangle^{\otimes b}$, one obtains the state $|\phi\rangle |0\rangle^{\otimes b} + |\perp\rangle$ using $\mathcal{O}(\text{polylog}(1/\varepsilon) + \text{poly}(pd))$ ancilla qubits and $\mathcal{O}(\text{polylog}(1/\varepsilon) + \text{poly}(pd))$ gates where

$$\left\| |\phi\rangle - \frac{1}{C_a \|f\|} \sum_{\mathbf{x}, \boldsymbol{\xi}} \check{a}(\mathbf{x}, \boldsymbol{\xi}) e^{2\pi i \mathbf{x} \cdot \boldsymbol{\xi}/P} \hat{f}(\boldsymbol{\xi}) |\mathbf{x}\rangle |\boldsymbol{\xi}\rangle \right\| < \frac{\varepsilon}{\sqrt{P^d} C_a}, \quad (8)$$

and $C_a \geq \sup |a| = \sup |\check{a}|$ is a constant. Here $|\phi\rangle$ denotes an unnormalized $2pd$ -qubit state.

Step 4. Sum over the frequency variable. Finally, after applying the Hadamard gate $H^{\otimes pd}$ to the $\boldsymbol{\xi}$ registers, we obtain the state

$$\left(I_{pd} \otimes |0\rangle^{\otimes pd} \right) \left(I_{pd} \otimes \langle 0|^{\otimes pd} \right) \left(I_{pd} \otimes H^{\otimes pd} \right) |\phi\rangle |0\rangle^{\otimes b} + |\perp'\rangle,$$

where

$$\left\| \left(I_{pd} \otimes |0\rangle^{\otimes pd} \right) \left(I_{pd} \otimes \langle 0|^{\otimes pd} \right) \left(I_{pd} \otimes H^{\otimes pd} \right) |\phi\rangle |0\rangle^{\otimes b} - \frac{1}{\sqrt{P^d} \|f\| C_a} \sum_{\mathbf{x}, \boldsymbol{\xi}} \check{a}(\mathbf{x}, \boldsymbol{\xi}) e^{2\pi i \mathbf{x} \cdot \boldsymbol{\xi}/P} \hat{f}(\boldsymbol{\xi}) |\mathbf{x}\rangle |0\rangle^{\otimes pd} |0\rangle^{\otimes b} \right\| < \frac{\varepsilon}{\sqrt{P^d} C_a},$$

which can be seen from (8). Therefore, one obtains the desired state on the $\mathbf{x}$ registers upon measuring $|0\rangle^{\otimes pd}$ for the $\boldsymbol{\xi}$ registers and $|0\rangle^{\otimes b}$ for the ancilla qubits. Notice that there is an extra scaling factor $\frac{1}{\sqrt{P^d}}$ due to the application of Hadamard gates, so the success probability is $\mathcal{O}(2^{-pd})$.

The block encoding scheme of the PDO (5) constructed in this section can be summarized in the following theorem:

Theorem 3.1. For a generic symbol $a(\mathbf{x}, \boldsymbol{\xi})$, a block encoding of the corresponding discretized PDO defined by (5) can be

$$\left(2^{\frac{pd}{2}} C_a, \mathcal{O}(\text{poly}(pd) + \text{polylog}(1/\varepsilon)), \varepsilon \right)$$

-block-encoded using the circuit displayed in Figure 4 with gate complexity

$$\mathcal{O}(\text{poly}(pd) + \text{polylog}(1/\varepsilon)),$$

where $C_a \geq \sup |a|$ is a constant.

4 Efficient block encoding for separable symbols

In this section, we are concerned with symbols with particular structures and an efficient block encoding of the corresponding PDOs with $\mathcal{O}(1)$ success probability is constructed. We first give the following definition for separable symbols.

Definition 4.1. A symbol $a(\mathbf{x}, \boldsymbol{\xi})$ is **separable** if $a(\mathbf{x}, \boldsymbol{\xi}) = \alpha(\mathbf{x})\beta(\boldsymbol{\xi})$.

We define

$$\check{\alpha}(\mathbf{x}) = \alpha\left(\frac{\mathbf{x}}{P}\right), \quad \check{\beta}(\boldsymbol{\xi}) = \beta\left(\boldsymbol{\xi} - P \sum_{\xi_j \geq P/2} \mathbf{e}_j\right), \quad (9)$$

where $\mathbf{e}_j$ is the j -th standard basis vector in $\mathbb{C}^d$. With help of the notations (9), the PDO (4) becomes

$$Af(\mathbf{x}) = \sum_{\boldsymbol{\xi} \in [P]^d} e^{2\pi i \mathbf{x} \cdot \boldsymbol{\xi}/P} \check{\alpha}(\mathbf{x}) \check{\beta}(\boldsymbol{\xi}) \hat{f}(\boldsymbol{\xi}), \quad \mathbf{x} \in [P]^d, \quad (10)$$

It is clear from the definition that $\check{\alpha}$ is P -periodic since α is 1-periodic, and we also have $\sup |\alpha| = \sup |\check{\alpha}|$, $\sup |\beta| = \sup |\check{\beta}|$.

For the PDO (10), we propose an efficient block encoding illustrated by the following circuit.

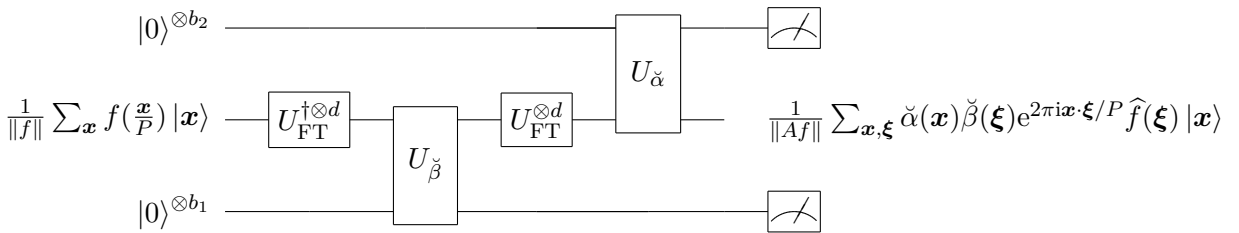


Figure 5: Circuit for an efficient block encoding for separable PDOs.

For the rest of this section, we explain the circuit in Figure 5 with more details.

Step 1. Apply QFT and multiply the factor $\check{\beta}(\boldsymbol{\xi})$. After applying the QFT gates $U_{\text{FT}}^{\dagger \otimes d}$, one obtains the state

$$\frac{\sqrt{P^d}}{\|f\|} \sum_{\boldsymbol{\xi}} \hat{f}(\boldsymbol{\xi}) |\boldsymbol{\xi}\rangle |0\rangle^{\otimes b_1 + b_2}.$$

Now that the input is represented on the frequency domain, one can naturally implement the multiplication of the factor $\check{\beta}(\boldsymbol{\xi})$. The corresponding block $U_{\check{\beta}}$ can be constructed using Proposition 2.2 with

$m = d, g = \check{\beta}$ and ε replaced by $\frac{\varepsilon}{2C_\alpha}$, where the constant $C_\alpha \geq \sup|\alpha|$. Then one obtains the state $(|\phi_1\rangle|0\rangle^{\otimes b_1} + |\perp_1\rangle)|0\rangle^{\otimes b_2}$, where $|\phi_1\rangle$ satisfies

$$\left\| |\phi_1\rangle - \frac{\sqrt{P^d}}{C_\beta \|f\|} \sum_{\xi} \check{\beta}(\xi) \hat{f}(\xi) |\xi\rangle \right\| < \frac{\varepsilon}{2C_\alpha C_\beta}, \quad (11)$$

using $\mathcal{O}(\text{poly}(pd) + \text{polylog}(1/\varepsilon))$ gates and $b_1 = \mathcal{O}(\text{poly}(pd) + \text{polylog}(1/\varepsilon))$ ancilla qubits. Here $C_\beta \geq \sup|\beta|$ is a constant.

Step 2. Apply QFT and multiply the factor $\check{\alpha}(x)$. In order to multiply the factor $\check{\alpha}(x)$ in the symbol, we apply QFT and convert the state to the space domain. Since

$$\sqrt{P^d} U_{\text{FT}}^{\otimes d} \sum_{\xi} \check{\beta}(\xi) \hat{f}(\xi) |\xi\rangle = \sum_{x, \xi} \check{\beta}(\xi) e^{2\pi i x \cdot \xi / P} \hat{f}(\xi) |x\rangle,$$

we have

$$\begin{aligned} & \left\| U_{\text{FT}}^{\otimes d} |\phi_1\rangle - \frac{1}{C_\beta \|f\|} \sum_{x, \xi} \check{\beta}(\xi) e^{2\pi i x \cdot \xi / P} \hat{f}(\xi) |x\rangle \right\| \\ &= \left\| U_{\text{FT}}^{\otimes d} |\phi_1\rangle - U_{\text{FT}}^{\otimes d} \frac{\sqrt{P^d}}{C_\beta \|f\|} \sum_{\xi} \check{\beta}(\xi) \hat{f}(\xi) |\xi\rangle \right\| \\ &= \left\| |\phi_1\rangle - \frac{\sqrt{P^d}}{C_\beta \|f\|} \sum_{\xi} \check{\beta}(\xi) \hat{f}(\xi) |\xi\rangle \right\| < \frac{\varepsilon}{2C_\alpha C_\beta}, \end{aligned} \quad (12)$$

where we have used (11) in the last line. By using [Proposition 2.2](#) again with $m = d, g = \check{\alpha}$ and ε replaced by $\frac{\varepsilon}{2C_\beta}$, the state $|\phi_1\rangle|0\rangle^{\otimes b_2}$ is mapped to $|\phi_2\rangle|0\rangle^{\otimes b_2} + |\perp_2\rangle$, where $|\phi_2\rangle$ satisfies

$$\left\| |\phi_2\rangle - \frac{1}{C_\alpha} D_{\check{\alpha}} U_{\text{FT}}^{\otimes d} |\phi_1\rangle \right\| < \frac{\varepsilon}{2C_\alpha C_\beta}.$$

In this step, $\mathcal{O}(\text{poly}(pd) + \text{polylog}(1/\varepsilon))$ gates and $b_2 = \mathcal{O}(\text{poly}(pd) + \text{polylog}(1/\varepsilon))$ ancilla qubits are used. Therefore, the final state is $|\phi_2\rangle|0\rangle^{\otimes b_1+b_2} + |\perp\rangle$, where $|\phi_2\rangle$ satisfies

$$\begin{aligned} & \left\| C_\alpha C_\beta |\phi_2\rangle - \frac{1}{\|f\|} \sum_{x, \xi} \check{\alpha}(x) \check{\beta}(\xi) e^{2\pi i x \cdot \xi / P} \hat{f}(\xi) |x\rangle \right\| \\ &\leq C_\alpha C_\beta \left\| |\phi_2\rangle - \frac{1}{C_\alpha} D_{\check{\alpha}} U_{\text{FT}}^{\otimes d} |\phi_1\rangle \right\| + C_\beta \left\| D_{\check{\alpha}} U_{\text{FT}}^{\otimes d} |\phi_1\rangle - \frac{1}{C_\beta \|f\|} \sum_{x, \xi} \check{\alpha}(x) \check{\beta}(\xi) e^{2\pi i x \cdot \xi / P} \hat{f}(\xi) |x\rangle \right\| \\ &< \frac{\varepsilon}{2} + C_\beta \left\| D_{\check{\alpha}} U_{\text{FT}}^{\otimes d} |\phi_1\rangle - D_{\check{\alpha}} \frac{1}{C_\beta \|f\|} \sum_{x, \xi} \check{\beta}(\xi) e^{2\pi i x \cdot \xi / P} \hat{f}(\xi) |x\rangle \right\| \\ &\leq \frac{\varepsilon}{2} + C_\beta \sup|\check{\alpha}| \left\| U_{\text{FT}}^{\otimes d} |\phi_1\rangle - \frac{1}{C_\beta \|f\|} \sum_{x, \xi} \check{\beta}(\xi) e^{2\pi i x \cdot \xi / P} \hat{f}(\xi) |x\rangle \right\| < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon, \end{aligned}$$

where we have used the inequality (12) and the fact that $C_\alpha \geq \sup|\alpha| = \sup|\check{\alpha}|$ in the last line.

By checking the definition of block encoding and adding up the gates and ancilla qubits used, we obtain the following theorem.

Theorem 4.2. If $a(\mathbf{x}, \boldsymbol{\xi}) = \alpha(\mathbf{x})\beta(\boldsymbol{\xi})$ is a separable symbol, then the discretized PDO (10) can be

$$(C_\alpha C_\beta, \mathcal{O}(\text{poly}(pd) + \text{polylog}(1/\varepsilon)), \varepsilon)$$

-block-encoded using the circuit displayed in Figure 5 with gate complexity

$$\mathcal{O}(\text{poly}(pd) + \text{polylog}(1/\varepsilon)),$$

where $C_\alpha, C_\beta > 0$ are constants such that $C_\alpha \geq \sup|\alpha|$ and $C_\beta \geq \sup|\beta|$.

Remark. In contrast to the result in Theorem 3.1, one can observe that the exponential factor $2^{\frac{pd}{2}}$ is removed, and thus the success probability for the circuit in Figure 5 is improved exponentially compared to the one in Figure 4.

5 Efficient block encoding for fully separable symbols with explicit circuits

For separable symbols, the circuit presented in Figure 5 significantly increases the success probability compared to the one in Figure 4. However, this relies on circuits for arithmetic functions, which can still be challenging to construct in practice. In this section, we develop a more explicit circuit with the help of QSP and QET. To begin with, we consider the fully separable symbols defined as follows.

Definition 5.1. A symbol $a(\mathbf{x}, \boldsymbol{\xi})$ is called **fully separable** if $a(\mathbf{x}, \boldsymbol{\xi}) = \alpha(\mathbf{x})\beta(\boldsymbol{\xi})$ with $\alpha(\mathbf{x}) = \alpha_1(x_1) \cdots \alpha_d(x_d)$ and $\beta(\boldsymbol{\xi}) = \beta_1(\xi_1) \cdots \beta_d(\xi_d)$ where each function in the set $\{\alpha_k\}_{1 \leq k \leq d} \cup \{\beta_k\}_{1 \leq k \leq d}$ is a real even function, a real odd function or an exponential function of the form $f(y) = e^{i\theta y}$ for some real parameter θ .

Similar with (9), we introduce the following notations:

$$\check{\alpha}_k(x_k) = \alpha_k\left(\frac{x_k}{P}\right), \quad \check{\beta}_k(\xi_k) = \beta_k\left(\xi_k - P \cdot \mathbb{1}_{\xi_k \geq \frac{P}{2}}\right), \quad 1 \leq k \leq d.$$

Then the discretized PDO (4) becomes

$$Af(\mathbf{x}) = \sum_{\boldsymbol{\xi} \in [P]^d} e^{2\pi i \mathbf{x} \cdot \boldsymbol{\xi} / P} \prod_{k=1}^d \check{\alpha}_k(x_k) \prod_{k=1}^d \check{\beta}_k(\xi_k) \hat{f}(\boldsymbol{\xi}), \quad \mathbf{x} \in [P]^d. \quad (13)$$

In order to block-encode the PDO (13), we adopt the following circuit, as shown in Figure 6, which is similar with the one in Figure 5:

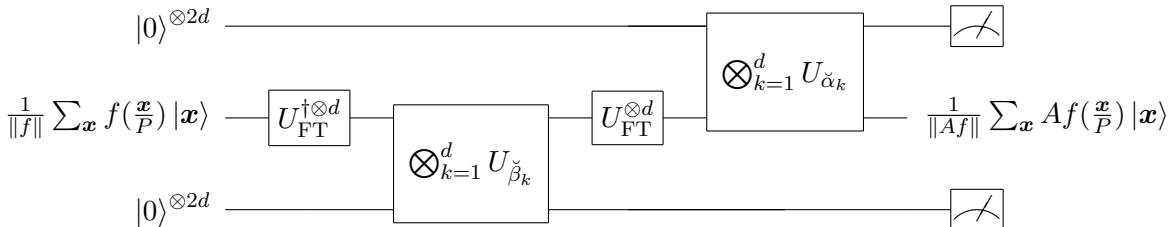


Figure 6: Circuit for explicit block encoding of fully separable PDOs.

Exploiting the fully separable structure of the symbol, one can construct explicit circuits for the diagonal multiplications shown in [Figure 6](#) by leveraging the QET subroutines prepared in [Section 2.3](#). Specifically, we define the overall polynomial degree approximation parameter:

$$\deg_a(\varepsilon) \equiv \sum_{k=1}^d (\deg_{\check{\alpha}_k}(\varepsilon) + \deg_{\check{\beta}_k}(\varepsilon)).$$

Denote by $U_{\check{\alpha}_k}$ and $U_{\check{\beta}_k}$ the block encodings of $\check{\alpha}_k(D_+)$ and $\check{\beta}_k(D_+) = \beta_k(D_-)$ obtained in [Proposition 2.4](#) with ε replaced by $\frac{\varepsilon}{2dC}$, respectively, where

$$C = \tilde{C}_\alpha \tilde{C}_\beta, \quad \tilde{C}_\alpha = \prod_{k=1}^d C_{\check{\alpha}_k}, \quad \tilde{C}_\beta = \prod_{k=1}^d C_{\check{\beta}_k},$$

and $C \geq \prod_{k=1}^d (\sup|\check{\alpha}_k| \sup|\check{\beta}_k|)$ since $C_{\check{\alpha}_k} \geq \sup|\check{\alpha}_k|$ and $C_{\check{\beta}_k} \geq \sup|\check{\beta}_k|$. Now we are ready to prove the following theorem that relies on the block encodings $\bigotimes_{k=1}^d U_{\check{\alpha}_k}$ and $\bigotimes_{k=1}^d U_{\check{\beta}_k}$ in [Figure 6](#).

Theorem 5.2. *If $a(\mathbf{x}, \boldsymbol{\xi}) = \alpha(\mathbf{x})\beta(\boldsymbol{\xi}) = \alpha_1(x_1) \cdots \alpha_d(x_d)\beta_1(\xi_1) \cdots \beta_d(\xi_d)$ is a fully separable symbol, then the corresponding PDO defined by (13) can be $(C, \mathcal{O}(d), \varepsilon)$ -block-encoded with gate complexity*

$$\mathcal{O}\left(pd \deg_a\left(\frac{\varepsilon}{2dC}\right) + p^2 d\right)$$

using the circuit displayed in [Figure 6](#).

Proof. Since each $U_{\check{\alpha}_k}$ is a $(C_{\check{\alpha}_k}, 2, \frac{\varepsilon}{2dC})$ -block-encoding for $\check{\alpha}_k(D_+)$ according to [Proposition 2.4](#), $\bigotimes_{k=1}^d U_{\check{\alpha}_k}$ is a $\left(\prod_{k=1}^d C_{\check{\alpha}_k}, 2d, \frac{\varepsilon}{2C_\alpha}\right)$ -block-encoding for

$$\bigotimes_{k=1}^d \check{\alpha}_k(D_+) = \bigotimes_{k=1}^d D_{\check{\alpha}_k} = D_{\check{\alpha}}.$$

Similarly, $\bigotimes_{k=1}^d U_{\check{\beta}_k}$ is a $\left(\prod_{k=1}^d C_{\check{\beta}_k}, 2d, \frac{\varepsilon}{2C_\beta}\right)$ -block-encoding for

$$\bigotimes_{k=1}^d \check{\beta}_k(D_+) = \bigotimes_{k=1}^d D_{\check{\beta}_k} = D_{\check{\beta}}.$$

Hence by the same argument as the proof of [Theorem 4.2](#), the circuit in [Figure 6](#) gives a $(C, 4d, \varepsilon)$ block encoding of the PDO (13) with gate complexity

$$\mathcal{O}\left(pd \deg_a\left(\frac{\varepsilon}{2dC}\right) + p^2 d\right),$$

where the $\mathcal{O}(p^2 d)$ term comes from the QFT part of the circuit. $\square$

6 Complexity Reduction via Approximate QFT for Fully Separable Symbols

As shown in [Theorem 5.2](#), the gate complexity for block encoding the fully separable PDO contains a term $\mathcal{O}(p^2 d)$, which originates from the exact QFT on d registers of p qubits. For a large grid size $P = 2^p$, this quadratic dependence on p can become the computational bottleneck. To further optimize the circuit, we can replace the exact QFT with the Approximate Quantum Fourier Transform (AQFT).

The exact QFT circuit on p qubits applies a series of controlled rotations. Let $R_j = R_z(\pi/2^{j-1})$. For a target qubit $j \in [p]$, the exact QFT applies controlled- R_{k-j} gates from all subsequent control qubits k (where $k > j$). The AQFT reduces the gate complexity by introducing a bandwidth parameter b and dropping all controlled- R_m gates where the distance $m = k - j > b$. So that the maximum number of controlled rotations acting on each qubit is bounded by b , reducing the gate complexity from $\mathcal{O}(p^2)$ to $\mathcal{O}(pb)$.

To rigorously bound the error introduced by this truncation, we first establish the error of the AQFT on a p -qubit register.

Lemma 6.1. *Let U_{FT} be the exact QFT operator on p qubits, and $\tilde{U}_{FT}^{(b)}$ be the AQFT operator with bandwidth parameter b , then*

$$\left\| U_{FT} - \tilde{U}_{FT}^{(b)} \right\| < \frac{2p\pi}{2^b}.$$

Proof. For any two quantum circuits composed of M unitary gates

$$U = U_M \cdots U_2 U_1, \quad V = V_M \cdots V_2 V_1,$$

we have

$$\|U - V\| = \|U_M \cdots U_2 U_1 - V_M \cdots V_2 V_1\| \leq \sum_{m=1}^M \|U_m - V_m\|.$$

The AQFT is constructed by replacing certain controlled- R_m gates in the exact QFT with the identity matrix I . The upper bound for the error of dropping a single controlled- R_m gate $C(R_m) = \text{diag}(1, 1, 1, e^{\pi i/2^{m-1}})$ is given by its spectral norm difference from the identity:

$$\|C(R_m) - I\| = \left| e^{\pi i/2^{m-1}} - 1 \right| = 2 \sin\left(\frac{\pi}{2^m}\right) \leq \frac{2\pi}{2^m}.$$

For a given target qubit j in the p -qubit AQFT circuit, we drop all controlled gates where the distance $m > b$. The total error accumulated on this single qubit wire is bounded by:

$$E_j \leq \sum_{m=b+1}^{p-j} \|C(R_m) - I\| < \sum_{m=b+1}^{+\infty} \frac{2\pi}{2^m} = \frac{2\pi}{2^b}.$$

Summing this error over all p target qubits in the circuit, the global spectral norm error of the AQFT is strictly bounded by:

$$\left\| U_{FT} - \tilde{U}_{FT}^{(b)} \right\| \leq \sum_{j=1}^p E_j < \sum_{j=1}^p \frac{2\pi}{2^b} = \frac{2p\pi}{2^b}.$$

□

With Lemma 6.1, we can now construct an improved block encoding for fully separable symbols by substituting $U_{FT}^{\otimes d}$ and $U_{FT}^{\dagger \otimes d}$ in Figure 6 with their approximate counterparts $\tilde{U}_{FT}^{(b)\otimes d}$ and $\tilde{U}_{FT}^{(b)\dagger \otimes d}$. This allows us to reduce the quadratic term $\mathcal{O}(p^2 d)$ to $\mathcal{O}\left(pd \log\left(\frac{pd}{\varepsilon}\right)\right)$ while preserving the desired overall accuracy.

Theorem 6.2. *If $a(\mathbf{x}, \boldsymbol{\xi}) = \alpha(\mathbf{x})\beta(\boldsymbol{\xi}) = \alpha_1(x_1) \cdots \alpha_d(x_d)\beta_1(\xi_1) \cdots \beta_d(\xi_d)$ is a fully separable symbol, then the corresponding PDO defined by (13) can be $(C, \mathcal{O}(d), \varepsilon)$ -block-encoded using the circuit displayed in Figure 6 with the exact QFT replaced by the AQFT. The gate complexity can be reduced to*

$$\mathcal{O}\left(pd \deg_a\left(\frac{\varepsilon}{4dC}\right) + pd \log\left(\frac{pd}{\varepsilon}\right)\right).$$

Proof. To maintain the overall precision ε for the block encoding, we allocate an error budget of $\frac{\varepsilon}{2}$ to the QET circuit approximations (diagonal multiplications) and $\frac{\varepsilon}{2}$ to the AQFT substitutions.

By the properties of the tensor product, the error of applying the AQFT independently across d dimensions is:

$$\left\| U_{\text{FT}}^{\otimes d} - \tilde{U}_{\text{FT}}^{(b)\otimes d} \right\| \leq d \left\| U_{\text{FT}} - \tilde{U}_{\text{FT}}^{(b)} \right\| < \frac{2pd\pi}{2^b}.$$

In the circuit shown in [Figure 6](#), the QFT and its inverse are applied exactly once. Hence, substituting both exact transforms with the AQFT introduces an additional error at most $\frac{4pd\pi}{2^b}$ to the final state. We require $\frac{4pd\pi}{2^b} \leq \frac{\varepsilon}{2}$, therefore, we can choose the bandwidth parameter

$$b = \left\lceil \log_2 \left(\frac{8pd\pi}{\varepsilon} \right) \right\rceil = \mathcal{O} \left(\log \left(\frac{pd}{\varepsilon} \right) \right).$$

With this choice of b , the gate complexity for each p -qubit AQFT is $\mathcal{O}(pb) = \mathcal{O} \left(p \log \left(\frac{pd}{\varepsilon} \right) \right)$. Since there are $2d$ such p -qubit transforms in total, the overall gate complexity contributed by the QFT is reduced to $\mathcal{O} \left(pd \log \left(\frac{pd}{\varepsilon} \right) \right)$.

Meanwhile, the error budget for the diagonal multiplications $U_{\check{\alpha}_k}$ and $U_{\check{\beta}_k}$ constructed via QET is tightened from ε to $\frac{\varepsilon}{2}$. Following the proof of [Theorem 5.2](#), this contributes $\mathcal{O} \left(pd \deg_a \left(\frac{\varepsilon}{4dC} \right) \right)$ to the gate complexity.

Adding the complexity of the QET and the AQFT together, the total gate complexity is:

$$\mathcal{O} \left(pd \deg_a \left(\frac{\varepsilon}{4dC} \right) + pd \log \left(\frac{pd}{\varepsilon} \right) \right).$$

Crucially, as explicitly derived in [Corollary 2.5](#), $\deg_a(\varepsilon) = \Omega \left(\log \left(\frac{1}{\varepsilon} \right) \right)$. Because the AQFT complexity term $\mathcal{O} \left(pd \log \left(\frac{pd}{\varepsilon} \right) \right)$ exhibits the same logarithmic scaling with respect to ε , it is asymptotically dominated by or absorbed into the QET cost. Omitting the constant factor $\frac{1}{2}$ in the error argument, the final gate complexity elegantly simplifies to:

$$\mathcal{O} \left(pd \deg_a \left(\frac{\varepsilon}{2dC} \right) \right).$$

This rigorous absorption confirms that replacing the exact QFT with the AQFT successfully removes the $\mathcal{O}(p^2d)$ bottleneck, while the overall complexity remains bounded exclusively by the cost of the QET. $\square$

7 Conclusion

In this paper, we presented a comprehensive study on the quantum block encoding of pseudo-differential operators (PDOs). We reconstructed the block encoding circuits for PDOs under different structural assumptions, ranging from generic symbols to separable and fully separable symbols, and provided detailed complexity analyses. For generic symbols, the block encoding is straightforward but suffers from an exponentially small success probability. By exploiting separable structures, the success probability can be exponentially improved.

Crucially, to address the $\mathcal{O}(p^2d)$ gate complexity bottleneck introduced by the exact Quantum Fourier Transform (QFT) in the fully separable case, we proposed a novel optimization utilizing the Approximate Quantum Fourier Transform (AQFT). By carefully allocating the error budget and rigorously bounding the error of the truncated controlled rotations, we successfully reduced the complexity of the Fourier transform steps to $\mathcal{O} \left(pd \log \left(\frac{pd}{\varepsilon} \right) \right)$. The asymptotic absorption analysis confirms that

the AQFT overhead is naturally dominated by the Quantum Eigenvalue Transformation (QET) cost, yielding an elegant and strictly bounded total complexity.

For future work, generalizing this AQFT optimization technique to the block encodings of other dense operators with implicit Fourier structures remains a promising direction.

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